Health Sciences - Graduate Level Assessment 13 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. What characteristic is not representative of a type IIb muscle fibre?
   (a) Low motor unit strength
   (b) Low glycogen storage capacity
   (c) High fatigue resistance
   (d) Low oxidative capacity
   (e) Low glycolytic capacity
2. Which of the following is NOT one of the major reasons for the increase in numbers of older adults?
   (a) Increases in life expectancy at a specified age
   (b) The increase in the rate of technological advancements
   (c) The decrease in childhood mortality rates
   (d) The steadily increasing birth rate
   (e) The Baby Boom generation
3. The synthesis of glucose from lactate, glycerol, or amino acids is called:
   (a) lipogenesis.
   (b) lipolysis.
   (c) lipo-genesis.
   (d) gluconeogenesis.
   (e) proteolysis.
4. The WHO made a strong recommendation that the maximum intake of free sugars by individuals within populations should be
   (a) 10-15%
   (b) >20%
   (c) <10%
   (d) >15%
   (e) <5%
5. Which of the following is not a non-coding RNA?
   (a) lncRNA
   (b) piRNA

(c) rRNA
(d) snRNA
(e) tRNA

True / False (10 marks)

Answer True or False

6. True or False: Proprioception pathways decussate in the corticospinal tract within the brain stem.
7. True or False: The karyotype 47,XX,+13 is associated with Down syndrome.
8. True or False: Medially directed strabismus is primarily caused by damage to the abducens nerve, which controls lateral eye movement.
9. True or False: Humans have the longest lifespan among the listed animals, often living beyond 80 years under optimal conditions.
10. True or False: Senescence is primarily characterized by the process of cellular division and growth.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of vitamin A in calcium absorption in the small intestine, analyzing its mechanisms and implications for overall health and nutrition.
12. Discuss the embryological development of the hyoid bone, including its origins from the pharyngeal arches and the implications for related anatomical structures.

End of Paper